

Internship Report: Retail Sales Tracker

Project: Day-8_SalesTracker_RetailSales — Google Sheets Dashboard

1. Executive Summary

Built a 3-layer retail dashboard in Google Sheets. The system transforms 1000 raw transactions into real-time KPIs, daily/weekly/monthly trends, and category analysis. Final audit: 100% accuracy with all 6 checks resulting in a precise MATCH.

2. Objective

- Create a scalable tracker for retail data.
- Ensure data integrity via master Lists.
- Build dynamic KPIs - Grand Total, Total Transactions, Avg Order Value, Pick-a-Date.
- Generate time-based rollups for trend analysis.

3. Workflow: Raw Data → Lists → Summary

3.1 Tab 1: Raw Data (RawEntries) - Foundation

Acts as the single source of truth containing 1000 rows of transaction data spanning from January 2023 to January 2024.

Formula applied: Total Amount = Quantity * Price per Unit

Format: Google Sheets Table (for dynamic expansion)

Example entries: 24/11/2023 - £150, 16/01/2023 - £2000, 13/12/2023 - £600.

3.2 Tab 2: Lists - Validation Master

Designed to prevent manual typos and ensure clean SUMIF logic across the dashboard. Implemented via *Data > Data Validation > List from a range*.

- **Categories:** Beauty, Clothing, Electronics
- **Genders:** Male, Female

3.3 Tab 3: Summary - The Dashboard

A. KPI Cards

GRAND TOTAL SALES £456,000.00 =SUM(Total Amount)	TOTAL TRANSACTIONS 1,000 =COUNTA(Transaction ID)
AVG ORDER VALUE (AOV) £456.00 =Grand Total / Transactions	PICK-A-DATE (INSTANT QUERY) £725 (on 13/12/2023) =SUMIF(Date, Selected Date, Amount)

B. Rollup Tables

Time-based aggregation built using combinations of UNIQUE, SUMIF, and COUNTIF.

- **Daily Rollup:** e.g., 01/01/23 £3,600 (3 txns), 06/02/23 £625 (4 txns).
- **Weekly Rollup:** Week Starting 26/12/22 £3,600 (3 txns). Peak Week: 30/01/23 £13,070 (25 txns).
- **Monthly Rollup:** Jan 23 £35,450 (76), Feb £44,060 (85), May £53,150 (105 - *Peak Month*), Oct £46,580 (96), Dec £44,690 (91).

C. Category Breakdown

Category	Total Volume	Percentage Share
Beauty	£143,515	31.47%
Clothing	£155,580	34.12%
Electronics	£156,905	34.41% (<i>Slight leader</i>)

Insight: The sales present a balanced portfolio with no single category dependency.

D. Audit Checks

All reconciliation checks returned a successful MATCH (Green), ensuring robust data integrity across the dashboard:

- ✓ **Sales on 2023-11-24 = £1,110:** Validates correct SUMIF logic.
- ✓ **Total Clothing = £155,580:** Validates manual data filter behavior.
- ✓ **Grand Total Match:** All Rollups precisely equal the Grand Total.
- ✓ **Daily Sales Match:** Validated against Grand Total and Row Count.
- ✓ **Weekly Sales Match:** Validated against Grand Total and Txn Count.

4. Formulas & Tools Used

SUM	SUMIF / SUMIFS	COUNTIF	AVERAGE	UNIQUE	SORT	WEEKNUM	DATE / TEXT
Data Validation	Conditional Formatting	Structured Table References					

5. Key Learnings

- **Architecture:** The importance of the 3-layer architecture (Raw > Lists > Report) as an industry best practice.
- **Reconciliation:** How to build audit-proof dashboards utilizing automated reconciliation checks.
- **Time Intelligence:** Recognized that time-intelligence modeling is absolutely crucial for retail analytics.
- **Data Hygiene:** Proved that strict data hygiene via validation lists prevents roughly 90% of downstream errors.
- **Dynamic Analytics:** The "Pick-a-Date" KPI serves as a highly powerful tool for ad-hoc management queries.

6. Conclusion

The completed tracker is fully validated and highly scalable to handle 10,000+ rows. It delivers actionable business insights—confirming £456k in volume, a £456 AOV, a peak month in May, and well-balanced performance across categories.

This assignment significantly strengthened my spreadsheet modeling, structural dashboarding, and core data integrity skills—all of which are essential competencies for a Data Analyst role.

Next Steps: Expand dashboard visual layers by adding charts, interactive slicers, and Google Apps Script for automated reporting.